

Number System Conversions — Quick Notes

Decimal • Binary • Octal • Hexadecimal — with solved examples

1. Decimal → Binary / Octal / Hex (Example: 25)

Target	Simple Steps	Answer
Binary	Keep dividing 25 by 2, note the remainder each time, stop at 0: $25 \div 2 = 12$, remainder 1 $12 \div 2 = 6$, remainder 0 $6 \div 2 = 3$, remainder 0 $3 \div 2 = 1$, remainder 1 $1 \div 2 = 0$, remainder 1 Now read remainders from last to first (bottom to top): 1 1 0 0 1	11001
Octal	Same idea, but divide by 8 instead of 2: $25 \div 8 = 3$, remainder 1 $3 \div 8 = 0$, remainder 3 Read bottom to top: 3 1	31
Hex	Divide by 16 instead: $25 \div 16 = 1$, remainder 9 Read bottom to top: 1 9	19

2. Binary → Decimal / Octal / Hex (Example: 101101)

Target	Simple Steps	Answer
Decimal	Write the value of 2 under each digit (1, 2, 4, 8, 16, 32...), starting from the right. Digits: 1 0 1 1 0 1 Values: 32 16 8 4 2 1 Add the values where the digit is 1: $32 + 8 + 4 + 1 = 45$	45
Octal	Split the binary number into groups of 3 digits, starting from the right: 101 101 Convert each group to one octal digit: $101 = 5$, $101 = 5$	55
Hex	Split the binary number into groups of 4 digits, starting from the right: 0010 1101 Convert each group to one hex digit: $0010 = 2$, $1101 = D$	2D

3. Octal → Decimal / Binary / Hex (Example: 57)

Target	Simple Steps	Answer
Decimal	Write the value of 8 under each digit (1, 8, 64...), starting from the right. Digits: 5 7 Values: 8 1 Multiply and add: $(5 \times 8) + (7 \times 1) = 40 + 7 = 47$	47
Binary	Turn each octal digit into its own 3-digit binary code: $5 = 101$, $7 = 111$ Join them together: 101 111	101111
Hex	First change to binary (see above): 101111 Then regroup into 4-digit blocks from the right: 0010 1111 Convert each block: $0010 = 2$, $1111 = F$	2F

4. Hex → Decimal / Binary / Octal (Example: 3A)

Target	Simple Steps	Answer
Decimal	Remember: hex letters A–F stand for 10–15, so A = 10. Write the value of 16 under each digit, starting from the right. Digits: 3 A(=10) Values: 16 1 Multiply and add: $(3 \times 16) + (10 \times 1) = 48 + 10 = 58$	58
Binary	Turn each hex digit into its own 4-digit binary code: $3 = 0011$, $A = 1010$ Join them together: 0011 1010	00111010
Octal	First change to binary (see above): 00111010 Then regroup into 3-digit blocks from the right: 000 111 010 Convert each block: $000 = 0$, $111 = 7$, $010 = 2$	072